

RESEARCH REPORT

Chronic inflammatory demyelinating polyneuropathy disease activity status: recommendations for clinical research standards and use in clinical practice

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Abstract Defining long-term outcomes in chronic inflammatory demyelinating polyneuropathy (CIPD) has been complicated by varying definitions of treatment response and differing scales measuring impairment or disability. An expert panel was convened to devise a CIPD Disease Activity Status (CDAS) and to classify long-term outcome by applying it to 106 patients with a consensus diagnosis of CIPD. Sixty of these cases were graded blindly by three independent reviewers to assess inter-rater reliability. The mean duration of follow-up was 6.4 years (range, 3 months–23 years). Eleven percent of patients were classified as cured (stable examination and off treatment for ≥ 5 years), 20% were in remission (stable and off treatment for < 5 years), 44% had stable active disease but required ongoing therapy for at least 1 year, 7% were improving after recent initiation of therapy, and 18% had unstable active disease (treatment naïve or treatment refractory). Excellent inter-rater reliability was observed (kappa scores: 0.93–0.97; $p < 0.0001$). The CDAS is considered a simple and reproducible tool to classify patients with CIPD according to disease activity and treatment status that can be applied easily in practice and potentially to select patients for clinical trials.

Key words: CIPD, disease activity, immune therapy, IVIg, neuropathy, treatment

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Introduction

Defining long-term treatment outcomes in patients with chronic inflammatory demyelinating polyneuropathy (CIPD) is complicated by varying definitions of

treatment response and differing outcome scales measuring impairment, functional problems at the disability and social levels, or impact on quality of life (Molenaar et al., 1999; van Nes et al., 2008; Hughes et al., 2010). The modified Rankin disability score has been frequently applied in retrospective studies (Simmons et al., 1995; Gorson et al., 1997; Tackenberg et al., 2007), perhaps due to its ease of application, but may be relatively insensitive to small but clinically meaningful functional improvement (Merkies et al., 2002). The Hughes scale was used in one long-term retrospective study (Kuwabara et al., 2006), but this scale was originally intended to be applied to patients with Guillain-Barré syndrome and is less sensitive than other scales because it does not reflect upper extremity function (Merkies et al., 2002). The Overall Disability Sum Score (ODSS), and the Overall Neuropathy Limitations scale derived from it are validated disability scales and are probably the best measure of disability but they have not been used to assess long-term outcome in CIDP (Merkies et al., 2002; Graham and Hughes, 2006). Furthermore, most clinical trials were short-term ones, assessing the efficacy of various therapies with endpoints after 6–12 weeks (Dyck et al., 1982; Hahn et al., 1996; Hughes et al., 2001; 2008; Mendell et al., 2001). Recent studies have demonstrated efficacy from intravenous immunoglobulin (IVIg) and high-dose, pulsed oral dexamethasone after 6-month and 1-year follow-up (Hughes et al., 2008; van Schaik et al., 2010), but data on long-term outcome (e.g., 5 or more years) are limited. In such cases a uniform grading system assessing disease activity in relation to treatment status would complement disability and impairment scales and is especially desirable. Such a grading system in CIDP patients would promote better classification of outcome after many years, on or off therapy, and perhaps encourage better patient selection for those with active disease for new treatment trials. The purpose of this study was to: (1) design and examine the reproducibility of a grading system to assess disease activity status relative to treatment status which can be used in clinical practice and research studies, and (2) classify long-term outcome by applying this classification scheme to the case descriptions of a well-defined cohort of CIDP patients.

Materials and Methods

Development of CIDP disease activity status

The CIDP disease activity status (CDAS) was created by an expert panel from the GBS/CIDP Foundation International Medical Advisory Board using a nominal group technique, and each iteration of the grading system was reviewed and discussed at national meetings

and by electronic mail exchanges among members of the panel until consensus was achieved, utilizing similar paradigms that have been effective for classifying disease status for patients with myasthenia gravis and other disorders (van der Heijde et al., 1990; Jones and Hunter, 1995; Jaretzki et al., 2000). These methods were modeled after a similar scale developed by the Myasthenia Gravis Foundation of America (MGFA, termed the MGFA Post-intervention Status) (Jaretzki et al., 2000). Points of discussion among investigators included: (1) can a classification system be developed that allows for a variable disease course and treatment response, and a variable duration of follow-up, and yet be simple and easy to apply; (2) does the concept of “cure” apply to patients with CIDP; (3) how might remission be defined, allowing for different evaluators in a community or academic environment; (4) how should those patients with persistent but stable long-term deficits be classified; and (5) how should stable patients requiring periodic treatment be classified. The following variables were considered relevant for the CDAS: (1) treatment status at the time of classification by the clinician (on or off therapy); (2) duration of treatment; (3) duration of follow-up (none to >5 years); (4) neurological examination at last follow-up, defined as normal or abnormal (excluding reflex abnormalities); and (5) treatment responsive or not, as defined by the judgment of the treating physician.

The CDAS specifically was constructed for easy application by practicing clinicians and researchers using broad categories of patient disease status, such as “cured,” “remission,” “stable active disease,” “improving,” or “unstable active disease.” These disease status categories are to be determined by the treating clinician. Patients are classified as cured if they had a stable neurological examination (subcategories of either normal or abnormal) and were off all treatment for 5 or more years. If patients are stable off treatment and the duration of follow-up off treatment was less than 5 years, they are classified as in remission (with a stable normal or abnormal examination). Patients are considered to have stable active disease if they require ongoing immunotherapy to maintain clinical stability for a year or more; in contrast, those who recently initiated therapy (for three months or more, but less than 1 year) and are responding to treatment are classified as improving. All other patients (treatment naïve, treatment for less than 3 months, or those with a progressive or relapsing-progressive course despite immune therapy of any duration) are considered to have unstable active disease (Fig. 1).

Application of CDAS

Once the classification system was developed and consensus was reached among panel participants,

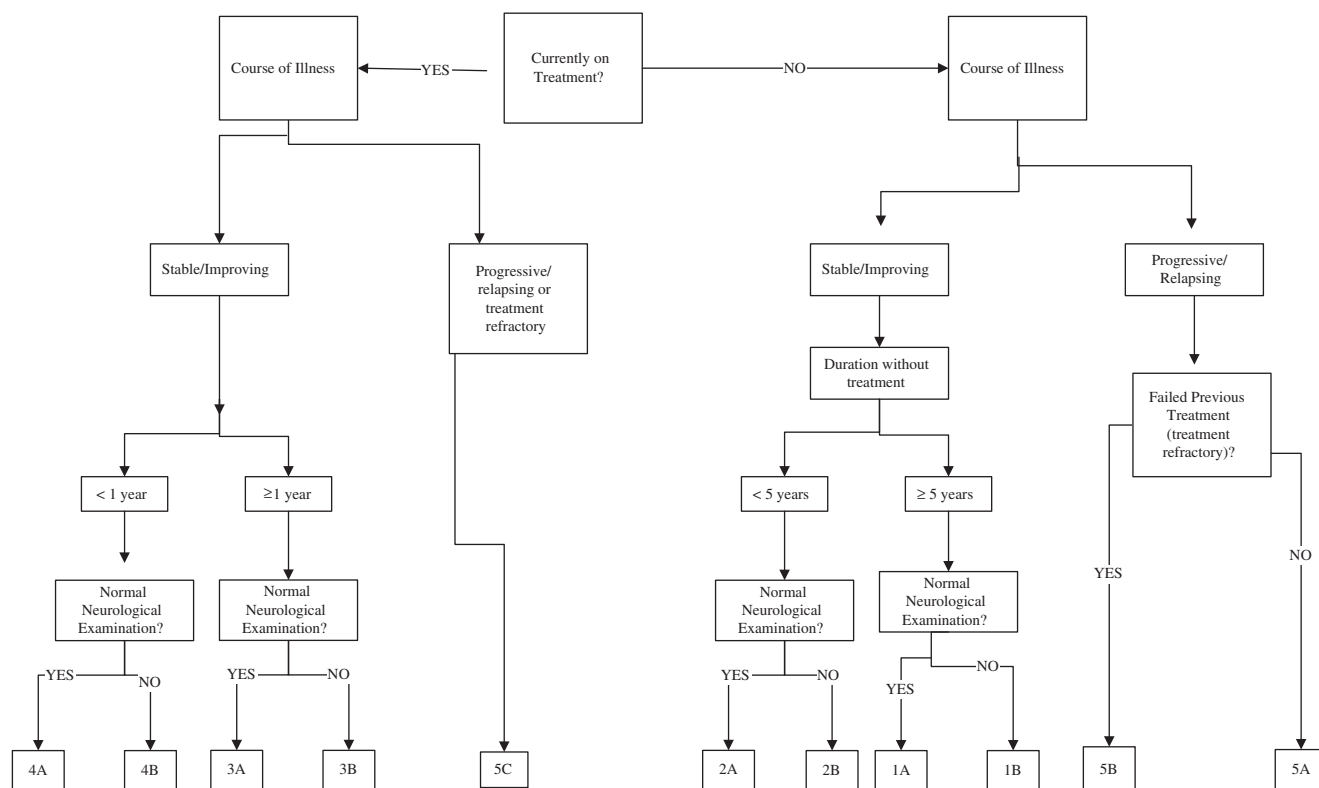


Figure 1. Algorithm to classify CIDP patients in relation to duration of disease, clinical response, neurological examination status, and duration of treatment.

the CDAS was applied to the case descriptions of a well-defined cohort of 106 patients with idiopathic CIDP. The case descriptions arose from an earlier study that has been previously described (Koski et al., 2009). Briefly, in the earlier study, the CIDP Criteria Working Group was formed to define and validate diagnostic criteria for CIDP using rigorous scientific methods. The group included 12 neurologists from the United States, Canada, and Europe, all of whom had expertise in the diagnosis and treatment of CIDP, and experts in epidemiology and biostatistics. A set of detailed case descriptions of patients with a chronic polyneuropathy were submitted for review by all the investigators, including information on the clinical features, laboratory testing, electrodiagnostic and cerebrospinal fluid studies, and nerve biopsy data when available. Each case description had three parts: (1) a structured patient profile completed by the submitting neurologist; (2) copies of the clinical notes and selected ancillary studies, including treatment responses described by the treating investigator; and (3) electrodiagnostic reports that included the primary electrophysiological data. Identifying personal information was deleted. All cases were reviewed by the participating neurologists who classified each case as: (1) CIDP; (2) chronic acquired demyelinating

neuropathy (CADP, e.g., multifocal motor neuropathy or paraprotein associated neuropathy); or (3) other neuropathy (e.g., due to diabetes mellitus, hereditary causes, toxin exposure, etc.). The diagnoses were provided via a web site that included a forum allowing online discussion of problematic issues that arose during the review process. For each case, the gold standard for case definition of CIDP, CADP, or other neuropathy was met if there was consensus on the diagnosis among at least 10 of 12 neurologists. If consensus was not reached after the initial round, the case was reposted for re-review, along with comments from the first review. If consensus was not reached after a second review, there was further informal consultation among reviewers by electronic mail or telephone. If there remained no consensus, the case was excluded. All decisions regarding treatment and assessments regarding clinical response were determined by the treating physician as documented in the case records. In contrast, the CDAS classification for these cases was made by the study investigators (see below).

To assess inter-rater reliability, 60 of the cases with a consensus diagnosis of CIDP from the previous study were classified using the CDAS independently by three investigators (K. C. G., I. N. vS., I. S. J. M.) who were blinded to the grading results of the others.

The remaining 46 cases were classified only by the principal investigator (K. C. G.).

Statistical analysis

Inter-rater reliability among the three raters was estimated using weighted kappa with confidence intervals (Cohen, 1968). A p value of <0.05 was considered statistically significant.

Results

Application of CDAS to a validated cohort of patients with idiopathic CIDP

Utilizing data from the previous CIDP Diagnostic Criteria Working Group Study (Koski et al., 2009), investigators submitted 346 cases, adequate information for review was available for 280, and consensus was achieved for 267 (95%) cases. A consensus diagnosis for idiopathic CIDP was established in 106 cases (Koski et al., 2009). A summary of the clinical features of consensus defined idiopathic CIDP cases from the previous CIDP Criteria Working Group study is provided in Table 1 and has been previously reported (Koski et al., 2009). The 106 consensus cases of CIDP had a mean duration of follow-up of 6.4 years (median, 5 years; range, 3 months–23 years); the results of CDAS scoring applied to this set of case descriptions are provided in Table 2. Sixty cases were independently reviewed by three investigators (K. C. G., I. S. J. M., and I. N. vS.) and scored; discordant cases were re-reviewed and discussed through electronic mail exchanges until there was full agreement and the final score was assigned. The remaining 46 cases were scored by one of the investigators (K. C. G.).

The concordance of the blinded application of the CDAS to the 60 case descriptions among any two of the three raters ranged between 87% and 90%; across the three raters the overall concordance was 87% with a weighted kappa of 0.84 ($p < 0.0001$). Excellent inter-observer reliability values were obtained between any two of the three raters; for K. C. G. vs. I. S. J. M., Kappa was 0.973 (95% confidence interval [C.I.], 0.936–0.986); for K. C. G. vs. I. N. vS., Kappa was 0.957 (95% C.I., 0.894–1.000); and for I. S. J. M. vs. I. N. vS., Kappa was 0.937 (95% C.I., 0.910–0.960).

Eleven percent of the patients were classified as cured, 20% were considered in remission, 44% had stable active disease on therapy, and 7% were improving. For those classified as cured, mean follow-up was 7.4 years (median, 8 years; range 5–12 years). Overall, 82% of treated patients had long-term stability off treatment, or they were stable or improving on treatment. In contrast, 18% were either untreated at the time of classification or refractory to previously administered immune therapy (Table 2).

Table 1. Clinical features of a consensus defined cohort of patients with CIDP.

Clinical Characteristics	Patients (n = 106)
Age (years)	
11–29	12 (11%)
30–59	63 (59%)
60+	31 (29%)
Male gender	56 (53%)
Caucasian	98 (92%)
History of spontaneous relapses	24 (23%)
Symmetric onset	88 (83%)
Multifocal onset	17 (16%)
Sensory symptoms	
None	6 (6%)
Asymmetric	3 (3%)
Both sides	97 (91%)
Facial weakness	16 (15%)
Symmetric on motor exam	94 (89%)
Distal weakness on exam	
No limbs	14 (13%)
1–3 limbs	22 (21%)
All four limbs	70 (66%)
Proximal weakness on exam	
None	29 (27%)
1–3 limbs	20 (19%)
All four limbs	57 (54%)
Number of absent/hypoactive reflexes	
None	3 (3%)
1–9	29 (27%)
All 10	74 (70%)
Abnormal sensory function	
No	9 (8%)
Asymmetric	2 (2%)
Symmetric	95 (90%)
Evidence of demyelination on biopsy	
No	1 (1%)
Yes	10 (9%)
Biopsy not done	95 (90%)
Evidence of axon loss on biopsy	
No	2 (2%)
Yes	6 (6%)
Not done	98 (92%)
CSF protein (mg/dl)	
≤45	10 (9%)
>45	73 (69%)
Not tested	23 (22%)
CSF cells (number/μl)	
<10	78 (74%)
≥10	5 (5%)
Not tested	23 (22%)

CIDP, chronic inflammatory demyelinating polyneuropathy; CSF, cerebrospinal fluid. Adapted from Koski et al. (2009).

Discussion

Although the clinical course of some CIDP patients extends over decades, no long-term assessment tool is uniformly applied to the outcome in these patients because: (1) most randomized controlled treatment trials are conducted only for weeks or months and do not report on prospective data regarding outcome after many years with this condition (Dyck et al., 1982; Hahn et al., 1996;

Table 2. CDAS classification scheme and proportion of CIPD patients within each category.

Patients (n = 106)		
1. Cure: ≥ 5 years off treatment		11% Cure
A. Normal examination	6 (6%)	
B. Abnormal examination, stable/improving	5 (5%)	
2. Remission: < 5 years off treatment		20% Remission
A. Normal examination	12 (11%)	
B. Abnormal examination, stable/improving	10 (9%)	
3. Stable active disease: ≥ 1 year, on treatment		44% Stable disease
A. Normal examination	18 (17%)	
B. Abnormal examination, stable/improving	29 (27%)	
4. Improvement: ≥ 3 months < 1 year, on Treatment		7% Improving
A. Normal examination	0 (0%)	
B. Abnormal examination, stable/improving	8 (7%)	
5. Unstable active disease: abnormal examination with progressive or relapsing course*		18% Active disease
A. Treatment naïve or < 3 months	6 (6%)	
B. Off treatment	7 (7%)	
C. On treatment	5 (5%)	

The classification is based upon the clinical assessment of the treating physician at the time of last evaluation. CDAS, CIPD disease activity status; CIPD, chronic inflammatory demyelinating polyneuropathy.

*5B and 5C refer to patients who were treatment refractory from prior therapy or worsening despite ongoing therapy.

Hughes et al., 2001; 2008; Mendell et al., 2001; van Schaik et al., 2010); (2) treatment responses are defined by using a variety of different measures reflecting different levels of impairment, disability, or quality of life; and (3) the definition of “improvement” has varied depending on the study (Dyck et al., 1982; Hahn et al., 1996; Hughes et al., 2001; 2008; 2010; Mendell et al., 2001; van Schaik et al., 2010). Recent studies have addressed the concept of “minimum clinically important difference” after treatment for patients with inflammatory neuropathies to identify appropriate outcome measures that are not only statistically significant but also clinically relevant (Merkies et al., 2010). A uniform grading system assessing long-term outcome in relation to treatment may help better classify overall disease status in patients with CIPD.

We have found that the CDAS is a simple and reliable scoring system that potentially could be applied successfully both in clinical practice and research studies. The CDAS is intuitive as patients move easily

from one category to another during follow-up and reassessment based upon: (1) treatment status; (2) duration on or off therapy; (3) stability of the neurological examination; and (4) response to treatment, as judged by the treating clinician (Fig. 1, Table 2). We found that the CDAS also has high inter-rater concordance reliability when applied by three independent and blinded investigators to a rigorously defined set of case descriptions of patients with idiopathic CIPD (Koski et al., 2009).

The concept of a “cure” in CIPD has not been addressed in prior longer-term outcome studies (Simmons et al., 1995; Gorson et al., 1997; Sghirlanzoni et al., 2000; Mygland et al., 2005; Kuwabara et al., 2006; Tackenberg et al., 2007). We applied a variation of the definition of cure that has been used as an outcome measure in oncologic practice, disease-free survival for 5 or more years off treatment. Using this paradigm, we defined cure in CIPD as stable disease off treatment for 5 or more years (CDAS 1A and 1B, Fig. 1), and observed that this occurred in 11% of our cohort; all had received immune therapy, and many had chronic static neurological deficits. This observation suggests that in a minority of patients, CIPD is a curable condition. However, immune disorders theoretically could relapse several years after a period of stability, and the notion that the cancer definition of cure may not necessarily apply to patients with CIPD should be considered. Nonetheless, the median duration of follow-up in the CIPD patients classified as cured CIPD was 8 years and likely allows a sufficient time for observation for a relapse.

We found that an additional 20% of our patients were considered in remission and off therapy for less than 5 years (CDAS 2A and 2B). Fourteen percent of the cases had fixed neurological deficits, yet were considered cured or in long-term remission by CDAS classification (CDAS 1B and 2B). This is most likely due to stable and irreversible axon loss which persists in those no longer receiving treatment and does not change with long-term follow-up (Bouchard et al., 1999). If this finding extends to similar CIPD patients in general practice, it suggests that CDAS 1B and 2B patients may be over-treated (Hughes et al., 2008; van Schaik et al., 2010). Our long-term observations suggest that further treatment may not be indicated in the setting of chronic stable deficits, and the potential side effects, cost, and patient inconvenience could be avoided (Sghirlanzoni et al., 2000; RMC Trial Group, 2009; Hughes et al., 2010). Furthermore, due to the varying duration of follow-up we could not determine from our data how many CIPD patients may have had a monophasic course and therefore may not have benefited from continued long-term immune treatment.

Just over half of the patients in this study (51%) required continued treatment (CDAS 3 or 4), usually IVlg, plasma exchange, corticosteroids, or various corticosteroid-sparing agents, and therefore were classified as “Stable Active Disease” or “Improving,” based upon the duration of therapy, because in the view of the treating physician treatment arrested further progression or improved impairment, functional disability, or quality of life. CDAS 3 and CDAS 4 categories included those with persistent neurological deficits or treatment-related fluctuations. It is noteworthy that almost one-third of patients had a normal neurological examination at the last follow-up (CDAS 1A, 2A, and 3A) regardless of treatment status at the time of case review. We found that 18% of our cohort were classified as CDAS 5, “Unstable Active Disease,” yet 6% were treatment naïve at the time of case review (CDAS 5A) and thus probably would be reclassified based upon a high likelihood of a favorable initial treatment response (Simmons et al., 1995; Hahn et al., 1996; Gorson et al., 1997; Sghirlanzoni et al., 2000; Hughes et al., 2001; Mendell et al., 2001; Mygland et al., 2005; Tackenberg et al., 2007). In contrast, 12% had severe disease with a progressive or relapsing course despite ongoing or prior therapy; 5% remained on treatment despite a lack of response (CDAS 5B), presumably to prevent further progression, but it was not clear from the case records that treatment was helpful. The remaining 7% had failed numerous treatments and were deemed treatment refractory by the treating clinician (CDAS 5C). With longer follow-up, those classified as CDAS 3 or 4 may be reclassified to CDAS 1 or 2 if they remain clinically stable and treatment can be discontinued, or CDAS 5 if there is clinical progression or a relapse despite therapy.

The available data on long-term outcomes in CIDP are surprisingly limited. Other investigators have reported a 26% “remission” rate at 5-year follow-up, although these patients were off treatment for only 2 years (Kuwabara et al., 2006). Both the prospective IVlg CIDP Efficacy (ICE) and pulsed high-dose dexamethasone versus standard prednisolone treatment for chronic inflammatory demyelinating polyradiculoneuropathy (PREDICT) trials have indicated that as many as 40%–55% of patients treated with periodic IVlg or pulse oral dexamethasone for 6 months remained in remission after 6 months off therapy, corresponding to the CDAS 2 category (Hughes et al., 2008; van Schaik et al., 2010). The CDAS could not be applied to the few other studies with longer follow-up data due to the lack of details regarding treatment status, examination findings, and duration of therapy. For example, in one small retrospective study with almost 9-year follow-up, 6 of 14 patients (43%) had minimal or no functional disability (ODSS 0-1), but there was

no mention of treatment status (Harbo et al., 2008). Other studies have referred to “long-term” outcomes with data limited only to 24-month follow-up (Simmons et al., 1995). In a report of 42 cases of idiopathic CIDP with a mean follow-up duration of 4.2 years, 13% were classified as being in complete remission off treatment (corresponding to CDAS 2) and 12% were still receiving therapy (CDAS 3) (Sghirlanzoni et al., 2000). In another study, 47 of 83 (56%) patients were considered “autonomous” by telephone follow-up using a modified Norris scale with an average duration of follow-up of 6 years, but there was no mention of treatment or examination status (Bouchard et al., 1999). Others reported that 8% of CIDP patients had minimal non-disabling symptoms, presumably off treatment, after 4-year follow-up (corresponding to CDAS 2) (Mygland et al., 2005). The disparity in the frequency of remission off therapy likely reflects heterogeneous patient selection (patients were often stratified as relapsing or chronic CIDP), different outcome measures, and lack of available data regarding the relationship of treatment status, duration of treatment, and duration of follow-up for individual cases. Similarly, retrospective reports found that 27%–39% of CIDP patients were still receiving immune modulating therapy after 4–5 years (Mygland et al., 2005; Kuwabara et al., 2006). Chronic plasma exchange and IVlg have been reported to be effective in maintaining improvement with follow-up as long as 5 years in some cases (Choudhary and Hughes, 1995). Others have reported response rates of 67%–82% after treatment with conventional immune therapies based upon 1 or more point improvement on the Rankin score (Gorson et al., 1997; Tackenberg et al., 2007), but in one study the duration of treatment and follow-up were not mentioned (Gorson et al., 1997).

Although the CDAS may be a potentially useful clinical outcome tool to assess the long-term efficacy of treatments for CIDP, it may have greater practical utility for patient recruitment in future clinical trials. The CDAS can be used to identify patients with long-term inactive disease off therapy (CDAS 1 and 2) and those who are treatment refractory (CDAS 5B and 5C), thus avoiding inappropriate patient selection. It has been recognized from recent treatment studies (Hughes et al., 2008; 2010; RMC Trial Group, 2009; van Schaik et al., 2010) that many CIDP patients are over-treated and have stable disease off therapy, and there have been placebo response rates as high as 40% (RMC Trial Group, 2009; Hughes et al., 2010). In contrast, those with CDAS 3, CDAS 4, and CDAS 5A would be ideal candidates for trial recruitment.

There are several important limitations in this study. Although rigorous clinical criteria were used to achieve a consensus diagnosis of CIDP, the case

material was reviewed retrospectively, and atypical CIDP cases were not considered. Furthermore, the fact that the treatment response was judged by the treating clinician may be criticized due to lack of strict definitions regarding improvement and inherent subjectivity. However, this type of overall assessment has been used successfully as a metric in treatment trials in the evaluation of other disorders (e.g., clinician global impression of change) (*Richter et al., 2005*), and the CDAS provides a systematic structure for determining disease status that clinicians perform routinely in practice. More importantly, the case descriptions reviewed in this study were of patients referred to and treated by specialists providing tertiary neuromuscular care, and thus the high frequency of patients classified as cured or in remission may be related to more aggressive treatment or other referral bias, and therefore may not reflect the experience of CIDP patients in the community treated by neurologists in general practice. Lastly, our preliminary observations regarding long-term outcomes should be interpreted cautiously because only half of our patients had follow-up for 5 or more years.

Further analysis of data from our validation study (*Koski et al., 2009*) may identify candidate variables associated with a favorable (categories 1–4) or unfavorable (category 5) prognosis. Long-term prospective follow-up is needed to further validate the utility of the CDAS, reduce the potential of selection bias, and confirm these initial observations.

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